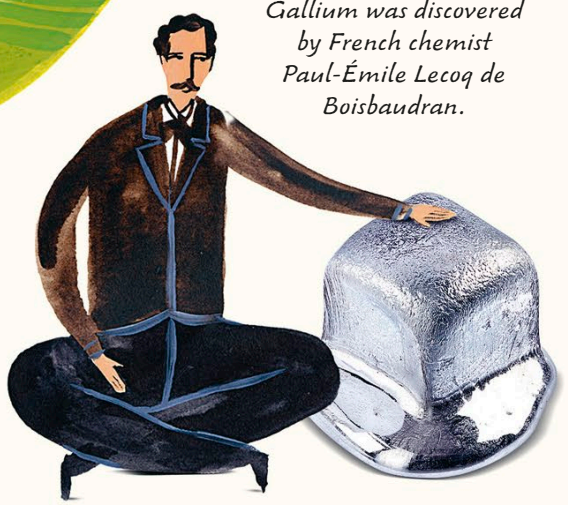
On the trip from Siberia to Moscow, and then St. Petersburg, Dmitri and his mother walked more than 1,000 miles (1,600 km)!



Gallium was discovered by French chemist Paul-Émile Lecoq de Boisbaudran.

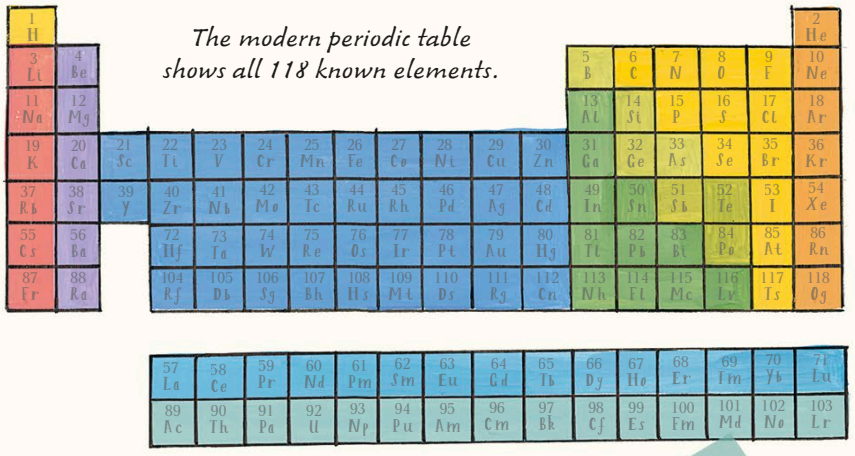
Predicting elements

From the gaps in his table, Dmitri was able to predict what undiscovered elements would be like. When the element gallium was discovered, he was proven right. It's a strange metal that is solid at room temperature but melts in your hands!

Elemental jigsaw puzzle

During the 1800s, dozens of new chemical elements—the building blocks of everything—were discovered. Dmitri collected information about each known element, such as the weight of their atoms, and tried to spot patterns. He noticed that elements of very different atomic weight could be similar in other ways.

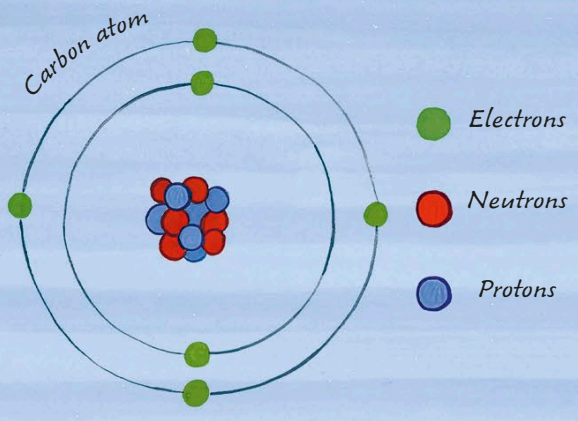
Dmitri made a chart that showed this pattern, leaving gaps where he thought new elements might be discovered. The elements were arranged in order of atomic weight, starting with the lightest. But he placed them in rows, or periods, so that elements with similar properties, such as metals, lined up to form groups. Today's periodic table of the elements looks different from Dmitri's first chart, but it's based on the same ingenious ideas. It is one of the most important tools in chemistry.



The modern periodic table shows all 118 known elements.



By smashing atoms together, physicists have been able to create elements that don't exist in nature. One was named Mendelevium, in honor of Dmitri.



Atomic structure

The atoms of each element have a set number of protons and electrons. The periodic table puts elements in order of this atomic number, beginning with hydrogen, whose atoms each have one proton and one electron. Carbon has six protons and electrons, so it is sixth on the periodic table.

